



3DEXPERIENCE®

ABAQUS POP产品介绍

DS DASSAULT
SYSTEMES | The 3DEXPERIENCE® Company

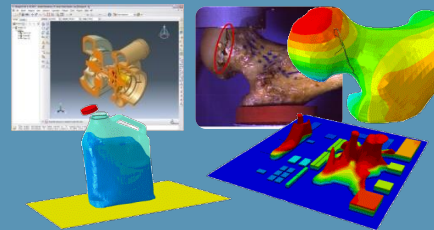


SIMULIA产品线及 ABAQUS POP组合包

SIMULIA ABAQUS结构产品组合-POP

Abaqus

- Abaqus/CAE
- Abaqus/Standard
- Abaqus/Explicit
- CEL, SPH, DEM, EMAG

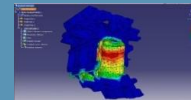


隐式与显式求解器

多物理场仿真

Fe-safe

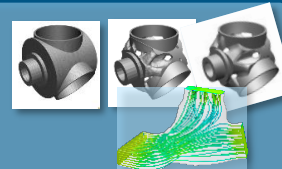
- 疲劳分析
- 耐久性预测



产品疲劳及
寿命分析

Tosca

- 结构与流体优化
- 概念及详细设计
- 拓扑、尺寸、形状、流道优化



拓扑优化

Isight

- 设计/运行时网关
- 应用程序组件
- 过程驱动程序



过程自动化和
参数优化

材料本构模型

AI/ML FOR ABAQUS

• Network-based Material Models

- Deep Material Network (DMN)

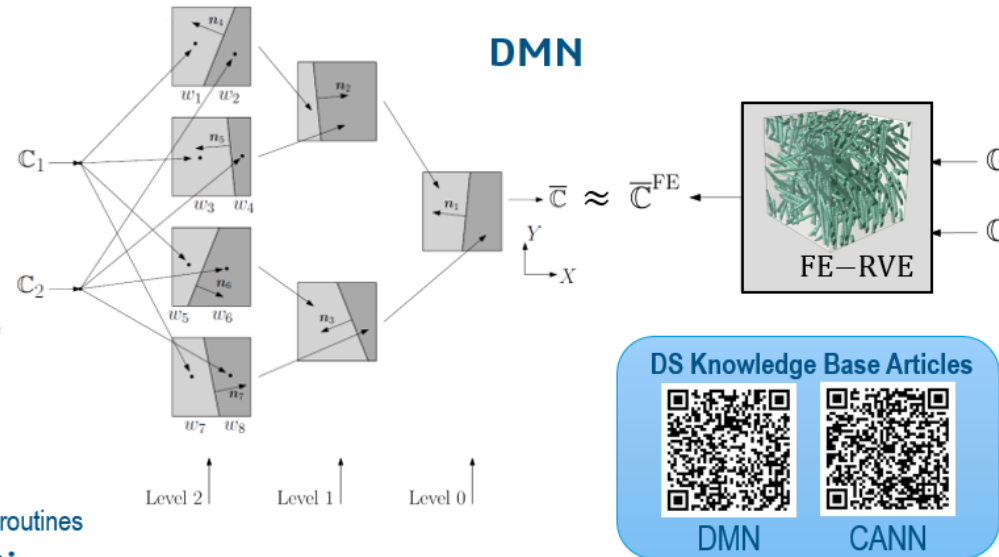
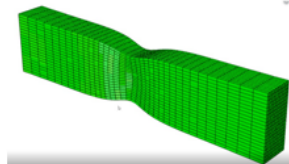
- › Network trained to match RVE response
- › Vs. Mean Field Homogenization
 - Reduce/eliminate constituent property calibration
 - Greater flexibility/fewer assumptions
 - Comparable or moderately higher computational expense
- › Microstructures – UD Fiber, Chopped Fiber
 - More in development

- Constitutive Artificial Neural Network (CANN)

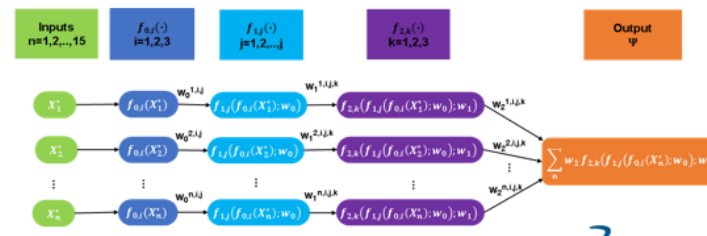
- › Network-based anisotropic hyperelastic potential
- DMN and CANN are implemented as built-in user subroutines

• Surrogate-Accelerated Material Calibration

- Feature in Material Calibration App
- Applies to **FE-Mode** calibrations
 - › Calibrating material using finite element model of specimen or part
- Surrogate dynamically trained during optimization and swapped in to accelerate objective evaluation



CANN



随着Abaqus R2025 HF03版本的发布，基于深度材料网络最新进展的复合材料本构模型已作为内置材料在Abaqus/Standard中推出。用户可通过该材料模型模拟纤维增强复合材料的力学响应。

ABAQUS POP包优势 与特点

Abaqus

- 强大的结构仿真求解器

线性 +
非线性仿
真

隐式+
显式求解
器

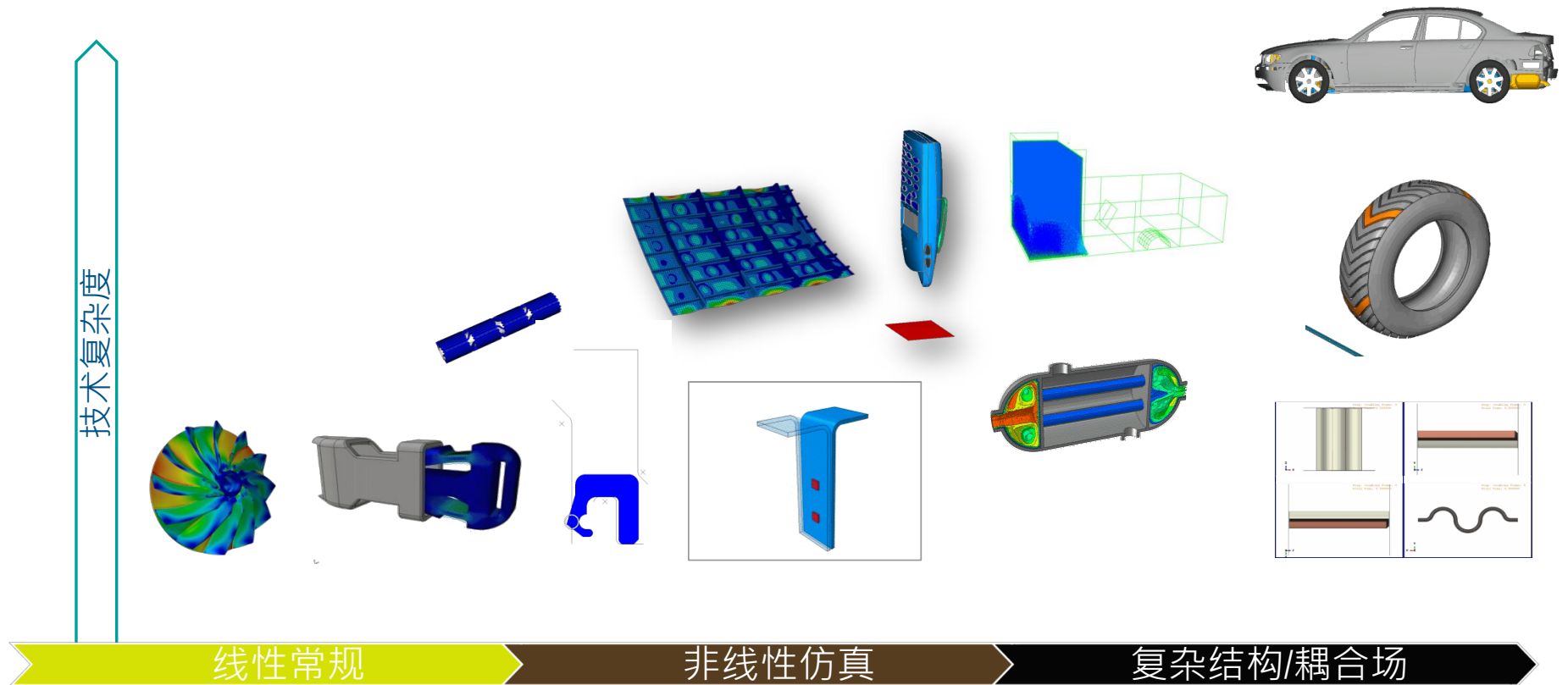
接触

断裂

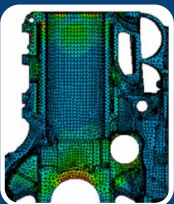
复合材料

联合仿真

SIMULIA ABAQUS

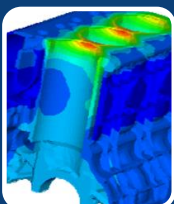


SIMULIA Abaqus



线性静力学、动力学和热传导

- 结构刚度强度
- 模态、振动分析
- 热分析



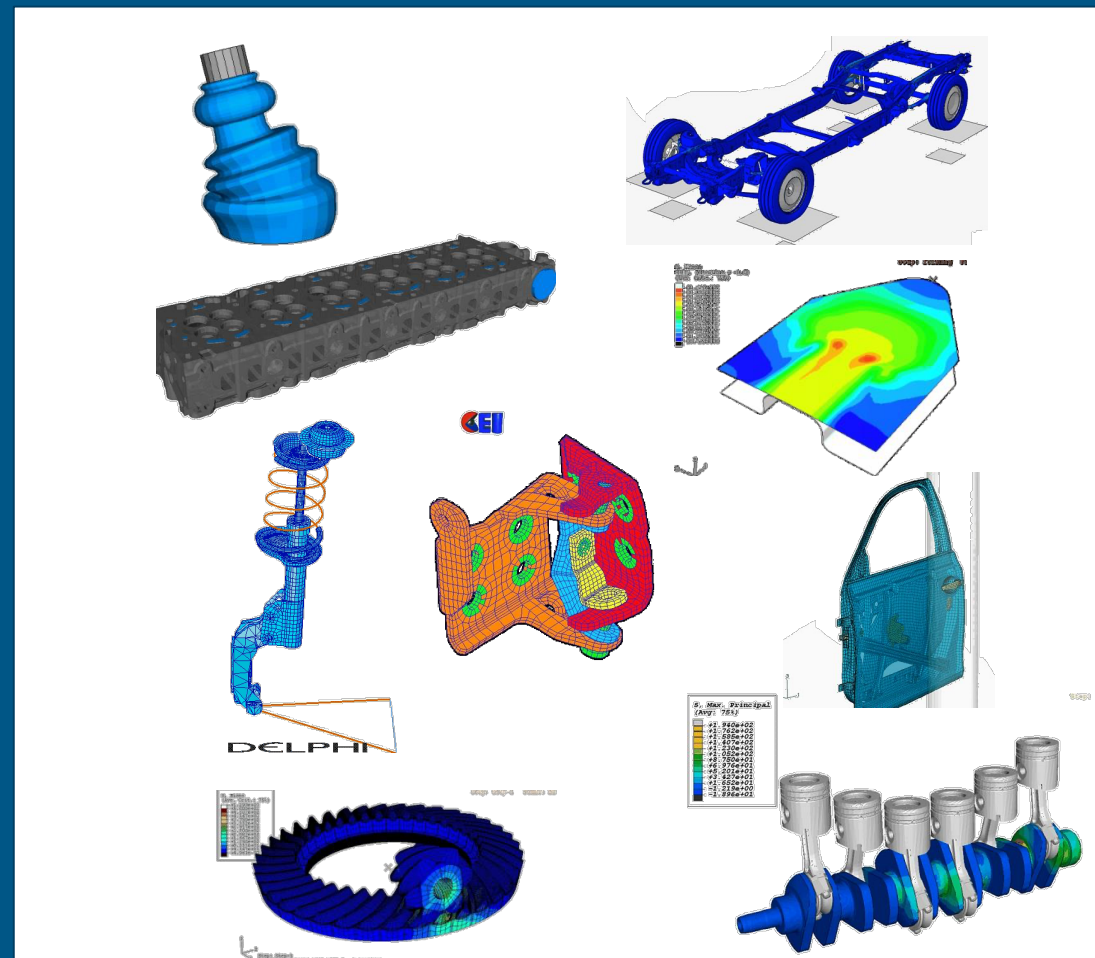
多物理场耦合分析

- 结构分析、热力分析
- 声学耦合分析
- SPH/CEL流固耦合



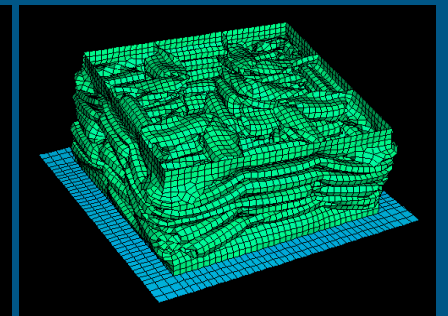
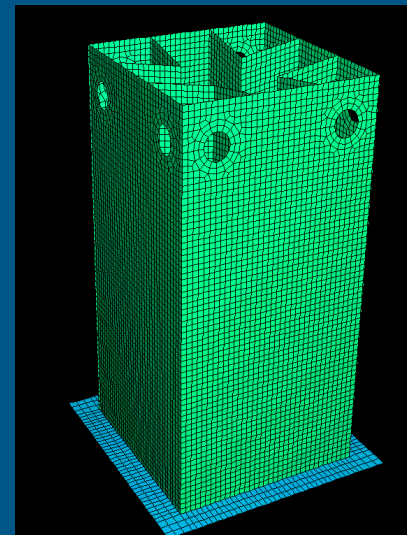
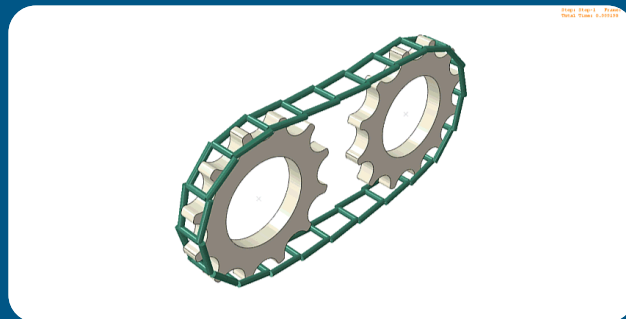
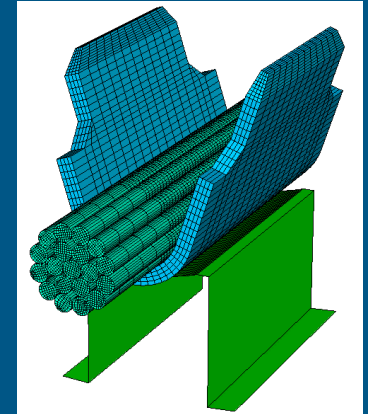
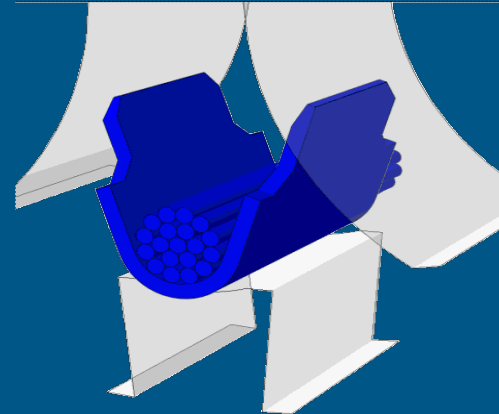
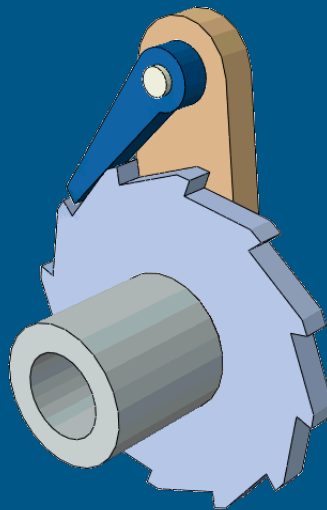
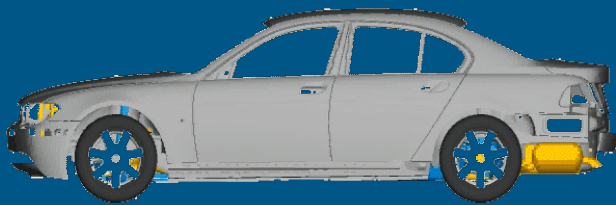
非线性和瞬态分析

- 接触分析
- 非线性材料、复合材料、裂纹、断裂、磨损、破坏
- 几何大变形、显式动力学分析



Abaqus核心能力—接触计算

- 通用接触
- 面-面接触
- 设置简便精度高



Courtesy of Alcan Mass

Abaqus核心能力—材料模型

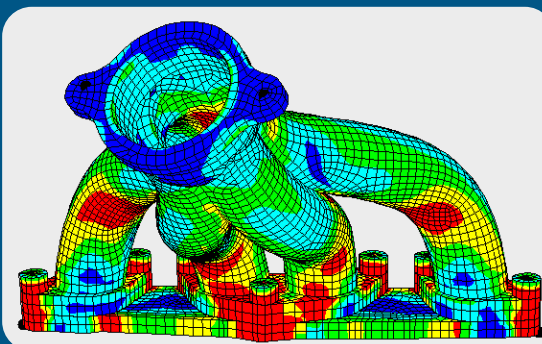
□金属、橡胶和复合材料

- 线性/非线性弹性和塑性
- 各向同性或各向异性
- 速率和温度的相关性

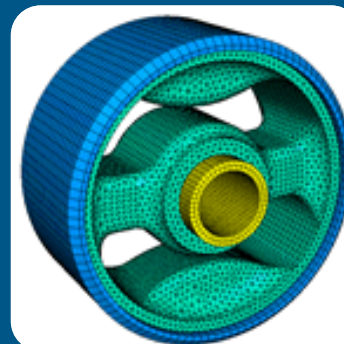
□其他材料包括

- 土壤和岩石
- 混凝土和陶瓷
- 浆料和聚合物
- 损伤和失效建模

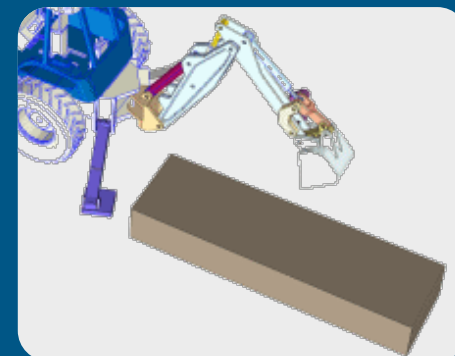
弹塑性铸铁



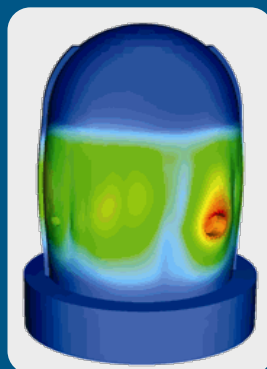
超弹性橡胶



土壤和岩石



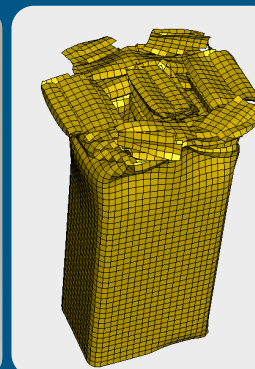
钢筋混凝土



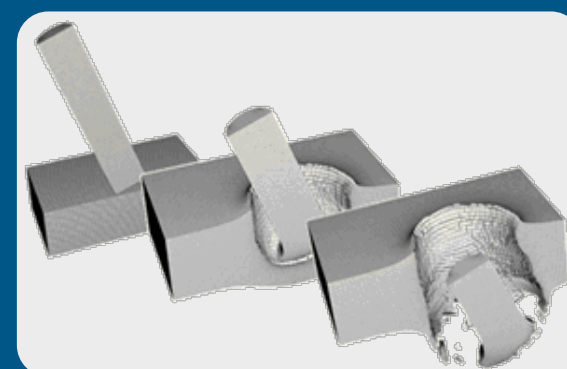
Test



Simulation



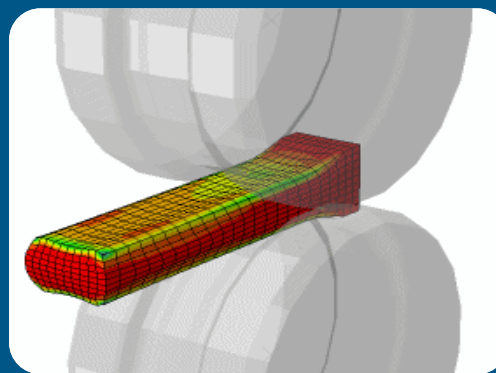
失效



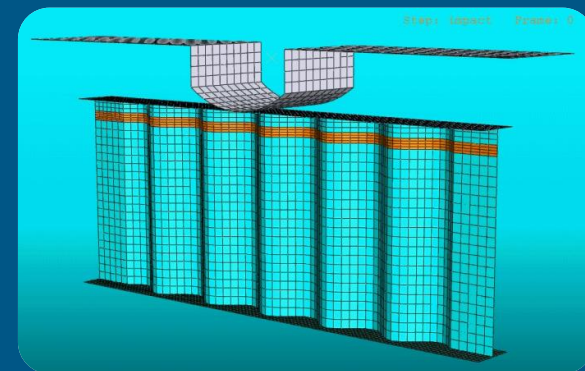
Abaqus核心能力—几何非线性

□大变形

- 刚度随着变形的发生而变化
- 大旋转，结构失稳-屈曲



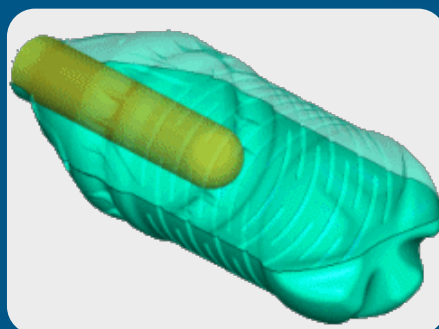
金属轧制



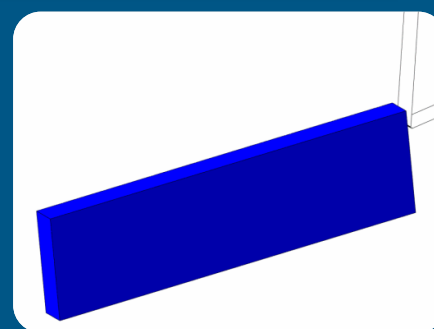
材料粉碎



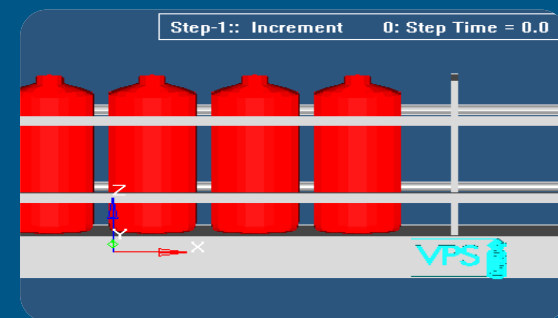
水瓶掉落



吹塑成型



切削



瓶子输送系统

ABAQUS 软件产品

Abaqus/CAE

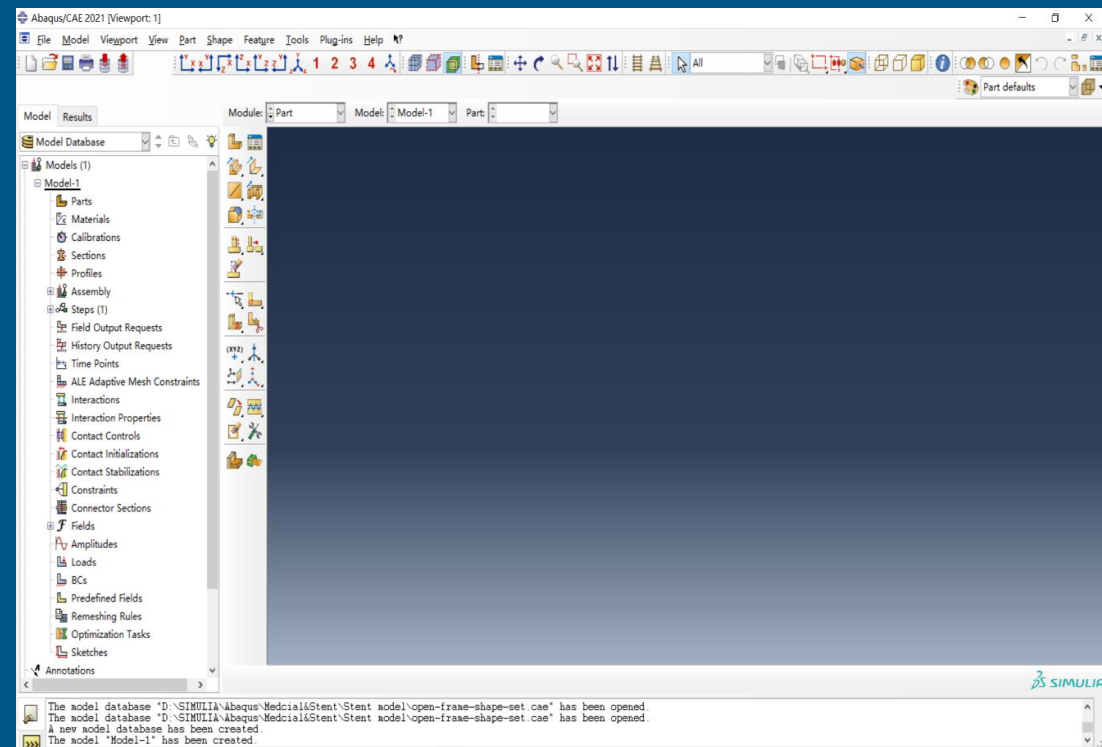
- 为Abaqus求解器提供快速交互式的前后处理环境
- Abaqus的建模、分析、监测和控制、以及结果评估的完整界面

Abaqus/Standard

- 主要用于结构静态、动态线性和非线性分析
- 耦合分析

Abaqus/Explicit

- 瞬态的大变形和高度非线性分析
- 可以在Abaqus/Standard分析结束状态进行继续分析



ABAQUS产品优势与特点

结构有限元和多物理场

- 线性和非线性
- 耦合场：声固耦合，热电耦合，热机耦合，流固耦合等
- 低频电磁场和SPH(smoothed particle hydrodynamics)

复杂材料模型

- 橡胶，塑料，粉末金属，人体组织，岩土，复合材料等等

可扩展性

- 用户子程序，Python二次开发，主流前后处理软件的接口

复杂和大型装配体

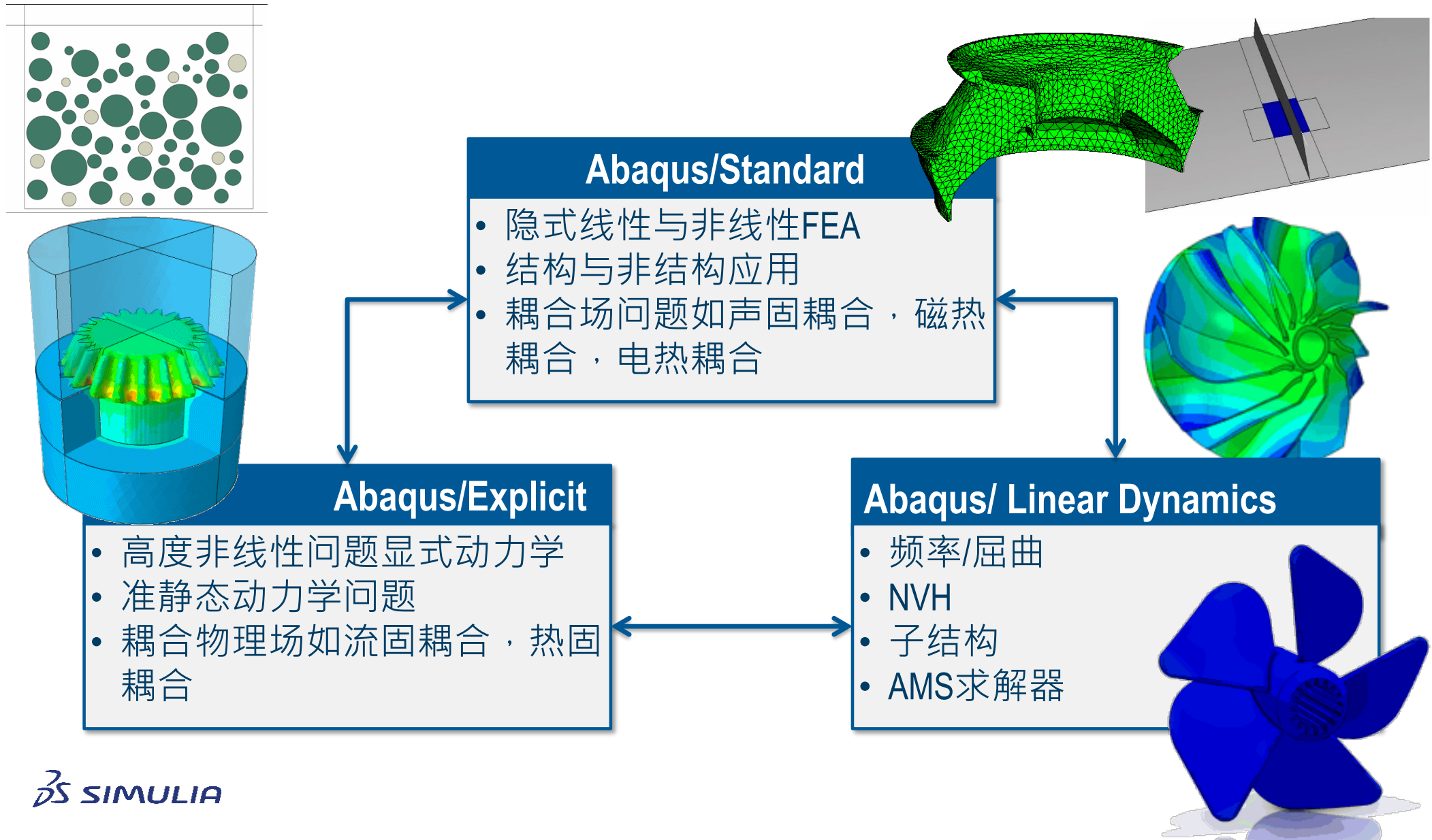
- 柔性多体动力学、控制和连接行为

接触，断裂和损伤失效

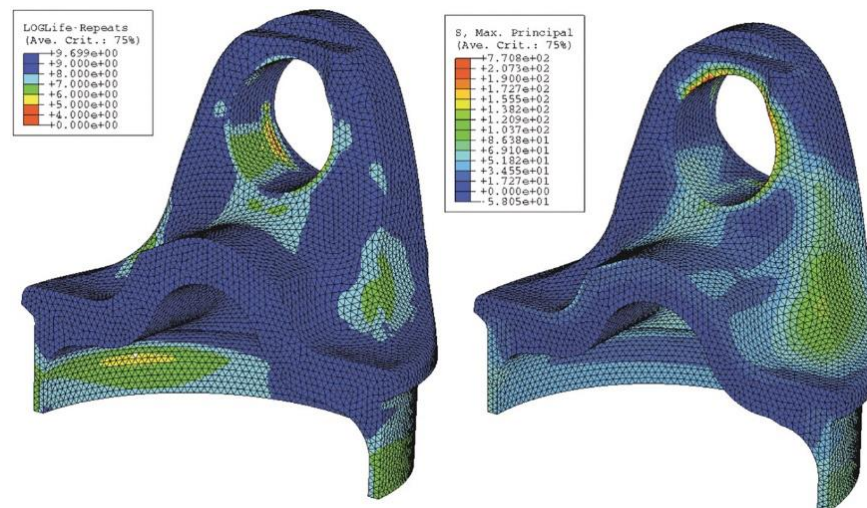
- 裂纹，冲击和碰撞

高性能计算

- 高效扩展以实现快速周转和大规模分析

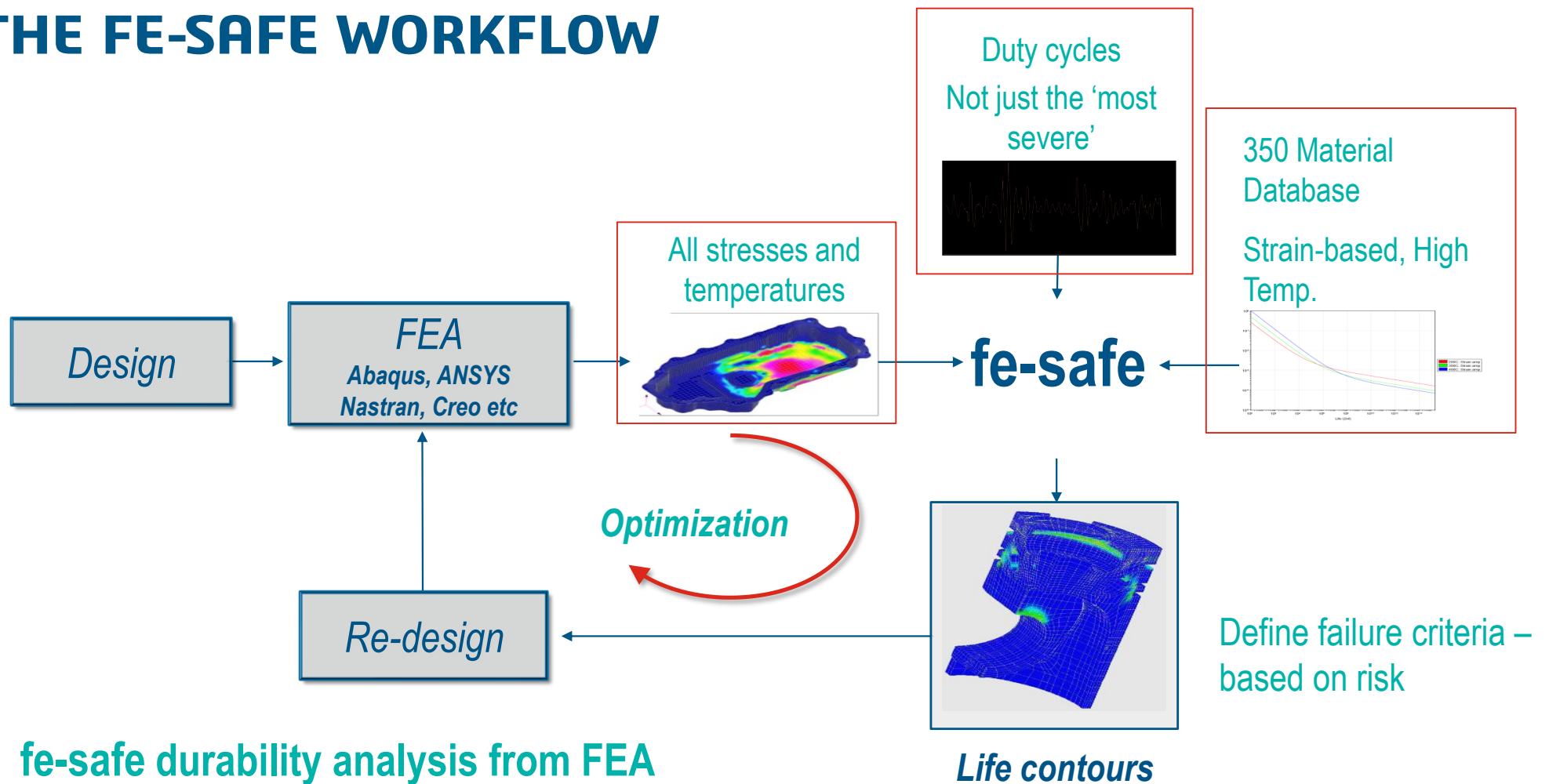


Fe-safe



Courtesy of Dana Automotive Systems, LLC

THE FE-SAFE WORKFLOW

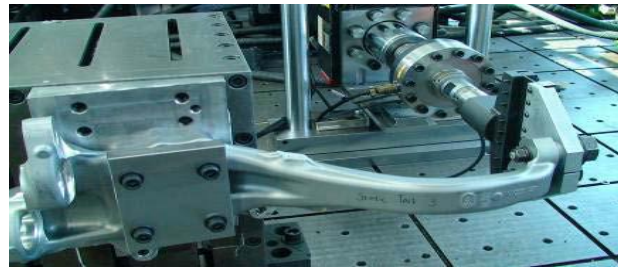
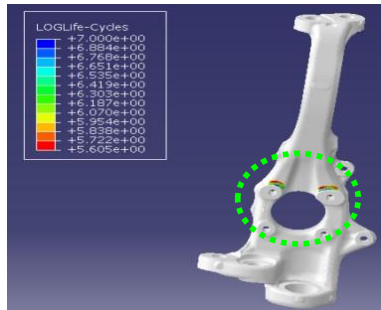
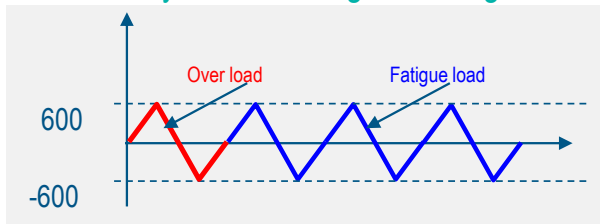


Why fe-safe?

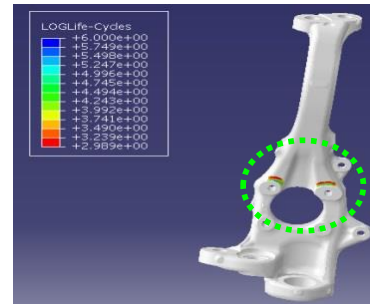
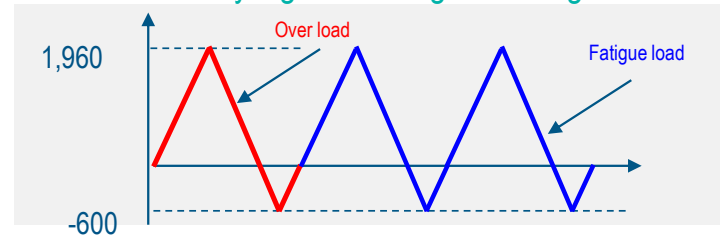
Testimonial: Hyundai Motor Company



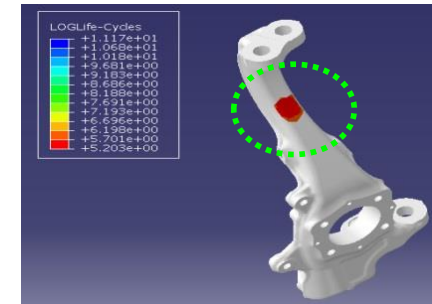
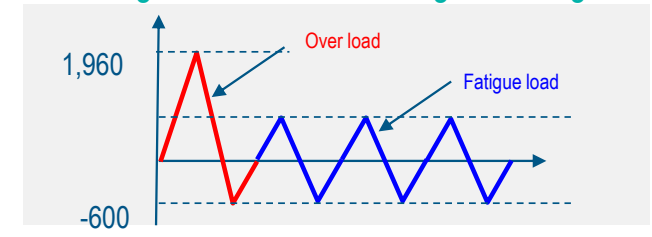
< only low level fatigue loading >



< only high level fatigue loading >



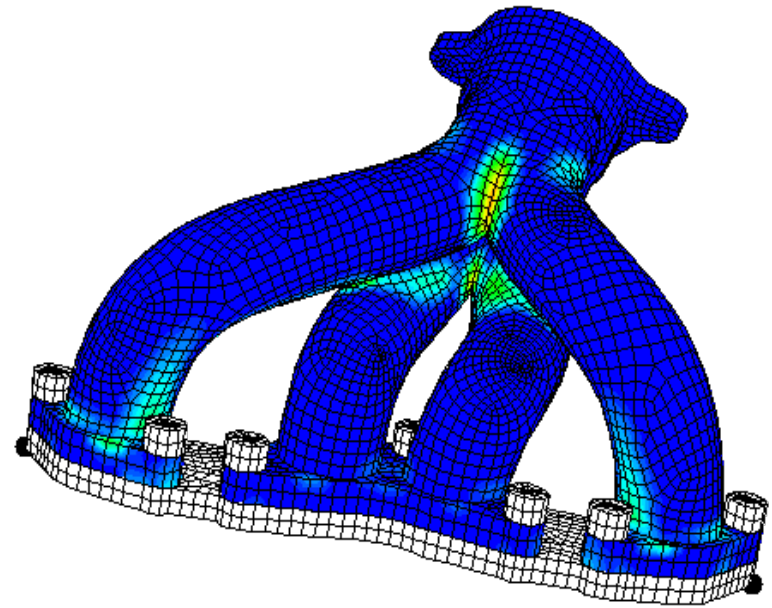
< high level + low level fatigue loading >



FE-SAFE疲劳仿真

专用模块

- ▶ Verity (焊缝)
- ▶ Rubber (橡胶)
- ▶ Composites (复合材料)
- ▶ TMF (热-结构疲劳分析包)
- ▶ TURBOlife (蠕变疲劳分析包)
- ▶ True-Load (试验数据校正)



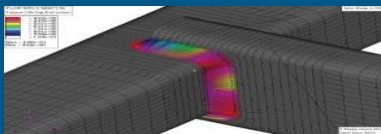
排气管热应力疲劳分析

专用模块-第三方工具

**Verity
Module in
fe-safe**

**Fatigue of Welded
Joints**

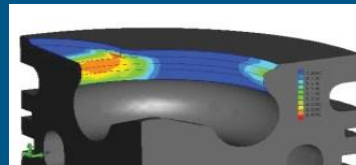
Battelle
The Business of Innovation



**fe-safe
/TURBOlife**

**Thermo-Mechanical
Fatigue**

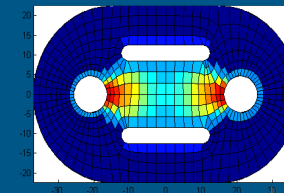
amec



**fe-safe
/Rubber**

Fatigue of Elastomers

EnduricaTM



KEY DRIVERS & DIFFERENTIATORS FOR FE-SAFE

综合性

- 先进、功能齐全的技术，具有许多独特的前沿功能
- 焊接疲劳、热-机疲劳和蠕变疲劳、橡胶疲劳的专业附加模块

准确性

- 先进的基于应变的多轴算法是 **fe-safe** 的核心
- 一致的客户反馈报告与测试结果具有极好的相关性

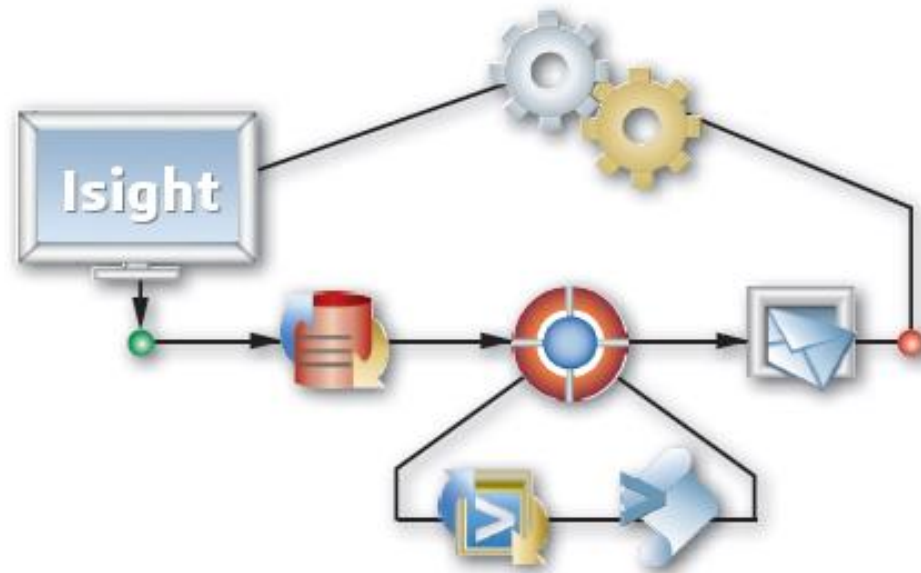
计算速度

- **Fe-safe** 计算效率高
- 分布式并行处理

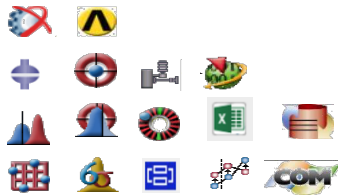
易用性和灵活性

- 许多默认设置和自定义设置
- 可保存标准分析，轻松设置批量分析
- **CMF** (自定义模块框架)：使用户能够将自己的专有方法插入 **Fe-safe**

Isight

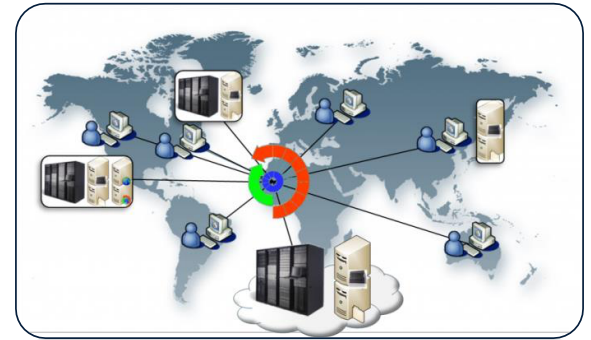


SIMULIA Automation & Optimization Technology



开放式组件框架

自动化流程执行、分发和并行化，监控执行

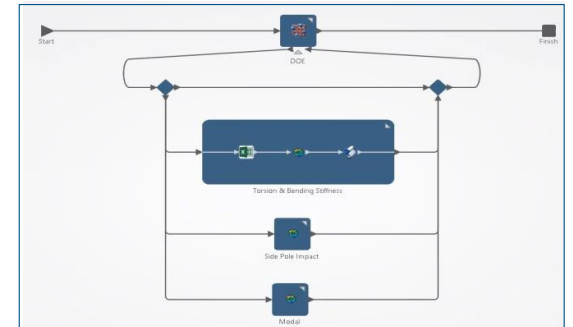


Isight 仿真过程自动化与设计探索



计算执行、结果可视化和后处理

仿真流程搭建



ISIGHT

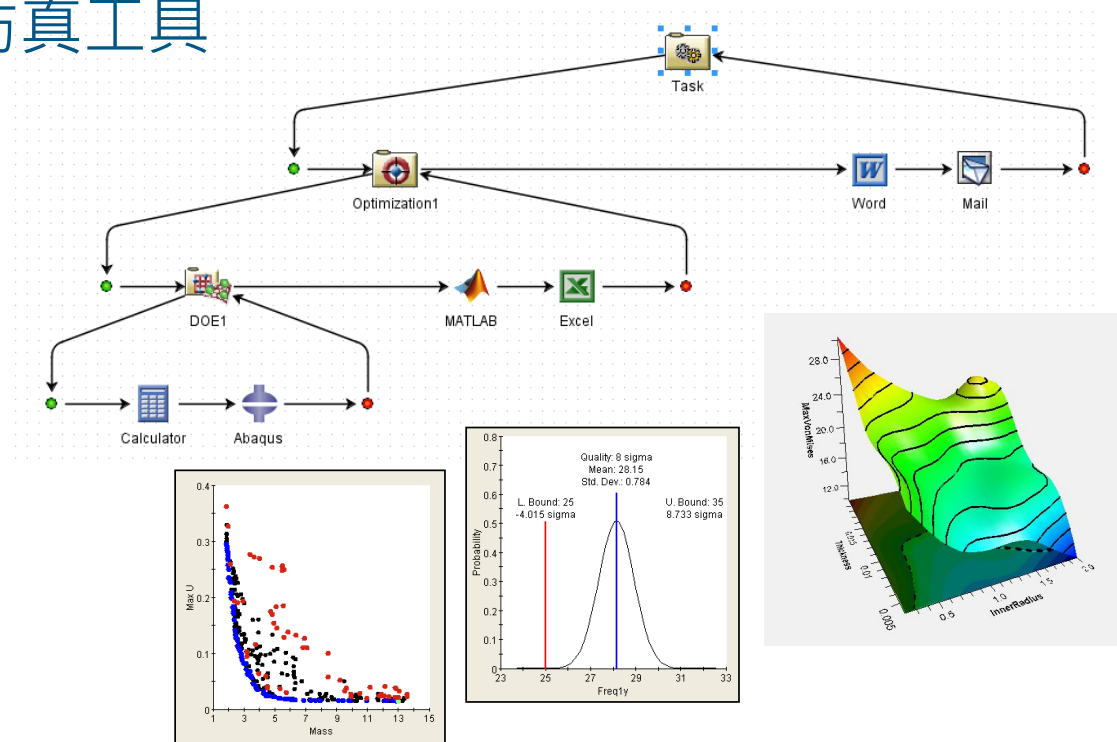
过程集成和设计优化 (PIDO) 解决方案

无需编程即可集成不同的仿真工具

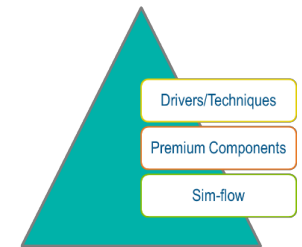
自动化仿真流程

使用先进技术优化设计

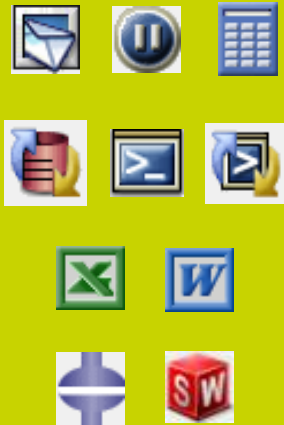
- ▶ DOE试验设计
- ▶ 优化
- ▶ 近似面
- ▶ 6 Sigma设计



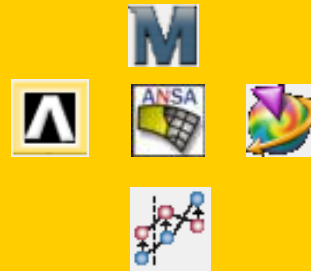
PREMIUM COMPONENTS



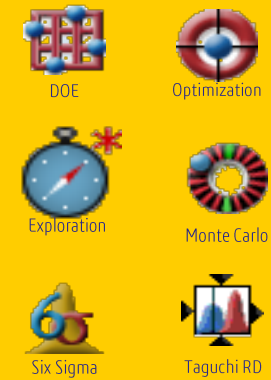
Core



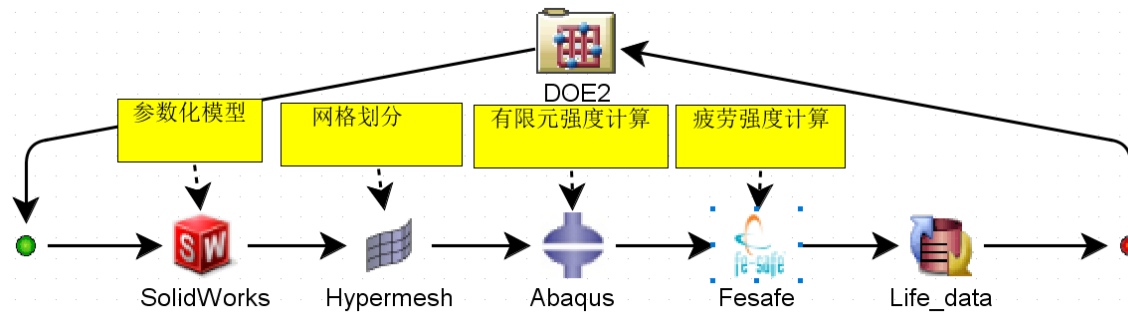
Premium



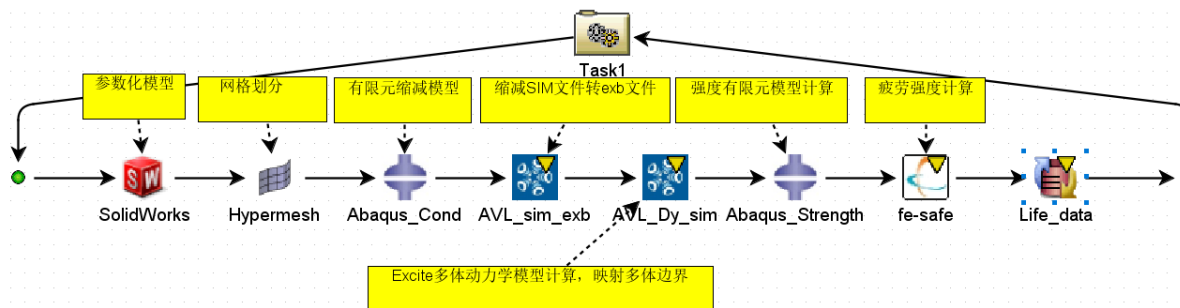
Premium



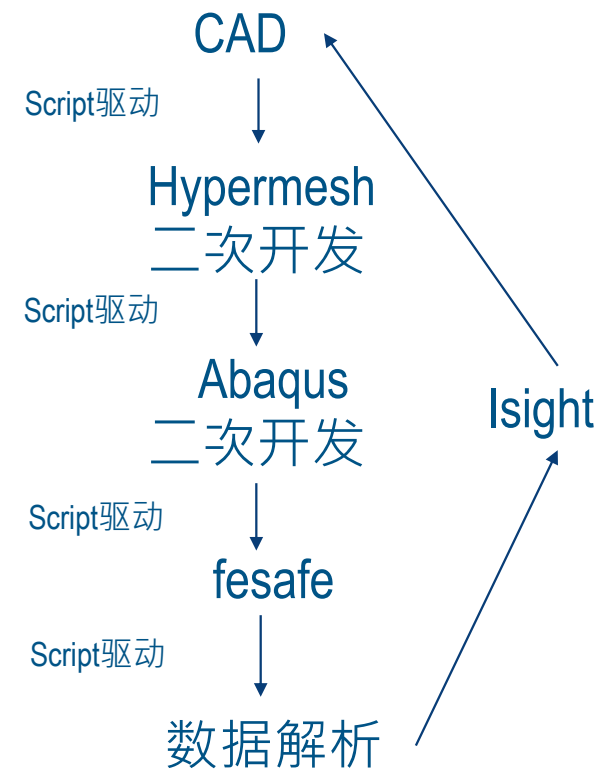
Isight仿真流程集成

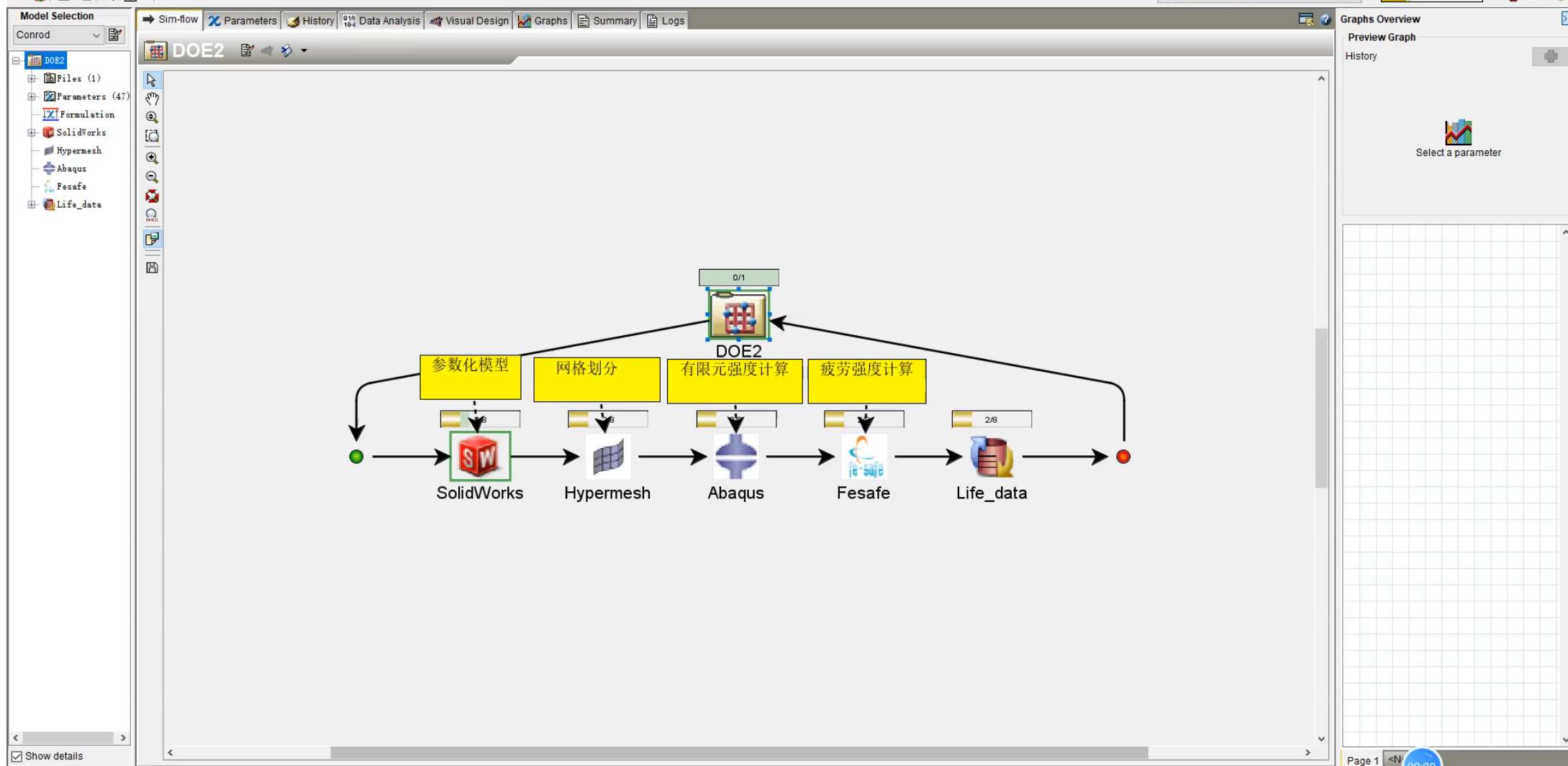


CAD、MESH、FEA、Fatigue自动化集成仿真



CAD、MESH、FEA、MBD、Fatigue自动化集成仿真





Graphs Overview

Preview Graph

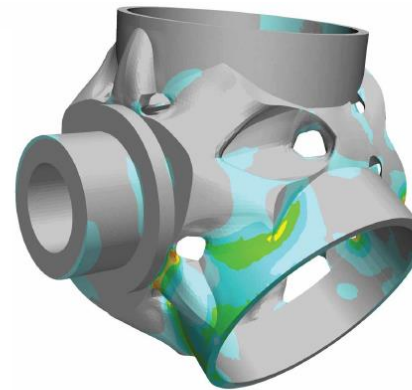
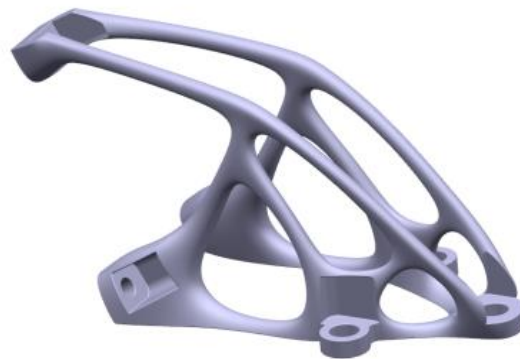
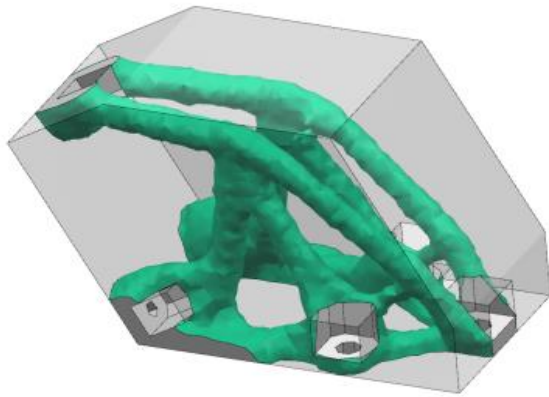
History

Select a parameter

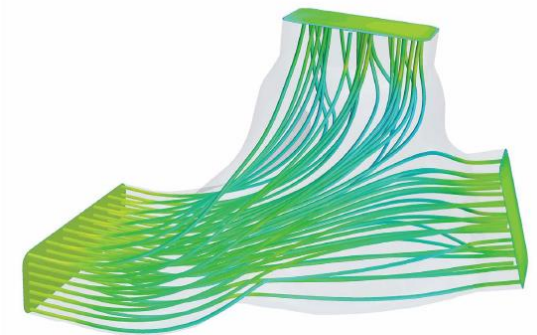
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Tosca Optimization Suite

基于 FEA 和 CFD 仿真的快速而强大的结构和流动优化解决方案



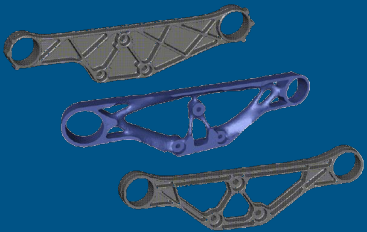
Tosca Structure
Optimized structural designs



Tosca Fluid
Optimized fluid flow
design concepts

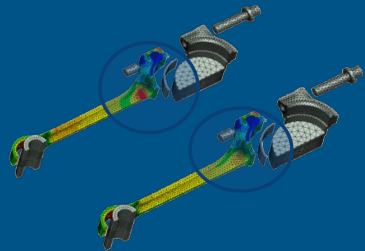
TOSCA 模块

Tosca Structure.topology



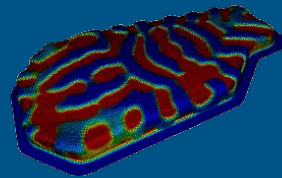
在设计空间生成一个最优机构，
满足重量最轻，刚度最大

Tosca Structure.shape



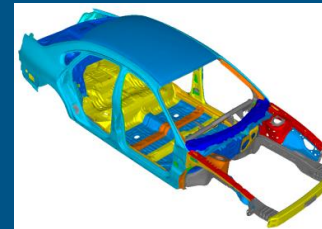
减小应力集中
提高疲劳寿命

Tosca Structure.bead



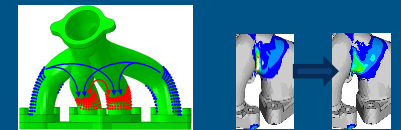
加强筋样式优化
提升刚度或模态

Tosca Structure.sizing



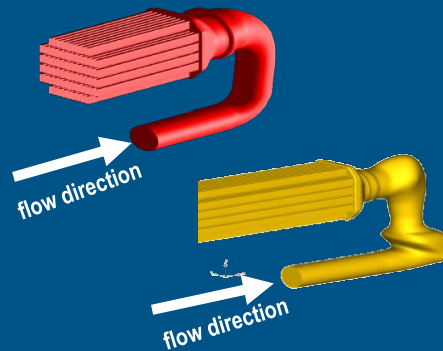
优化复杂板壳结构
提升刚度或模态

Tosca Structure.morph



通过控制变动网格
自动生成或评价设计变化

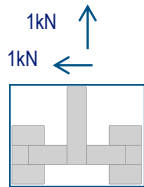
Tosca Fluid.topology



通道流量拓扑优化以降低压降

Fluent
STAR-CCM+

Comparison of Nonlinear Contact and tied Constraint



Stress Analysis

Arrangement of Material and stress analysis

Eliminating material that does not contribute in stress analysis

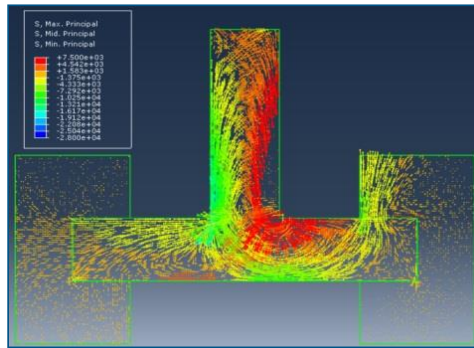


Fig 1 Nonlinear Contact

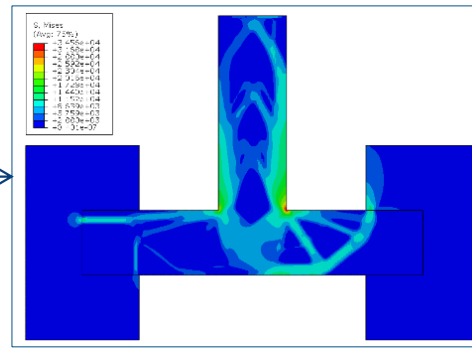


Fig 2 Von Mises Stress distribution in Optimization cycle 67

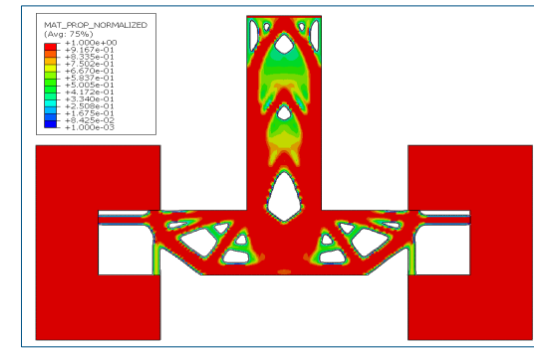


Fig 3 Iso density plot at 0.3 in cycle 67

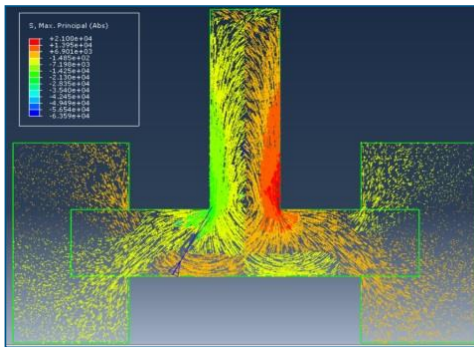


Fig 4 Tied Constraint

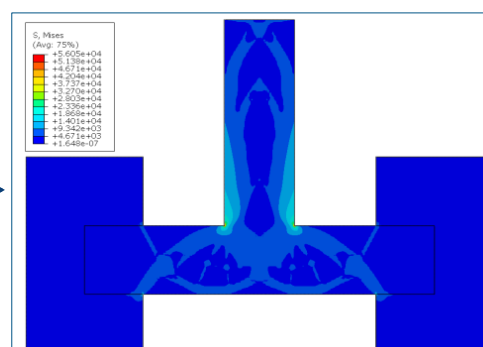


Fig 5 Von Mises Stress distribution in Optimization cycle 35

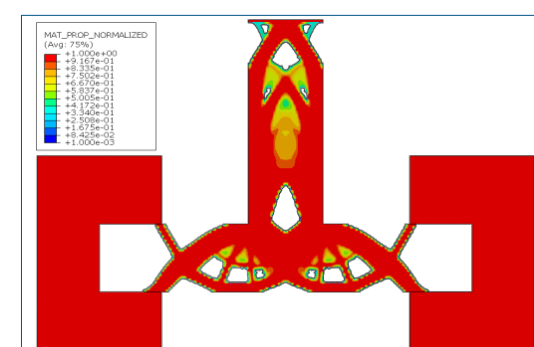


Fig 6 Iso density plot at 0.3 in cycle 35

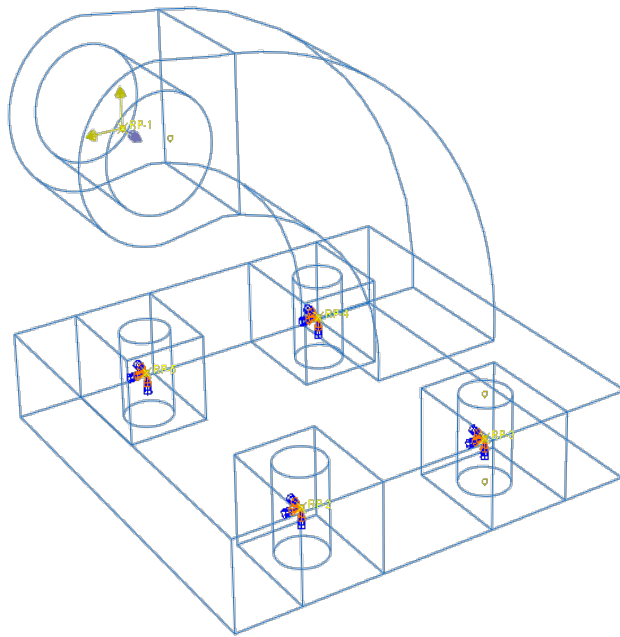


Fig 1 Loading and Boundary Conditions

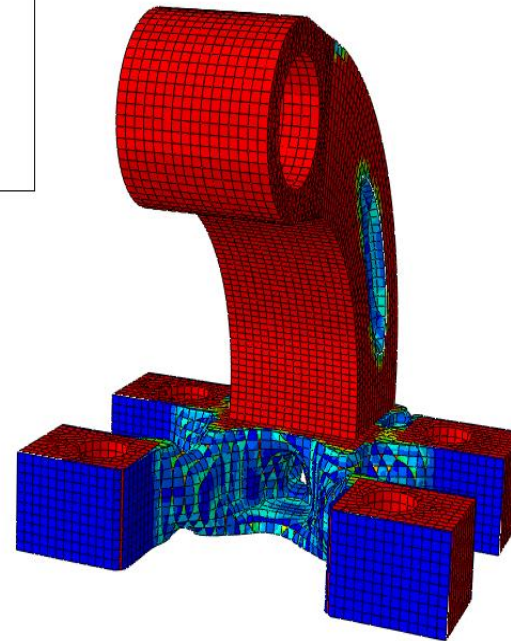
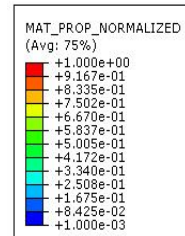


Fig 2 Only bolt hole regions are frozen

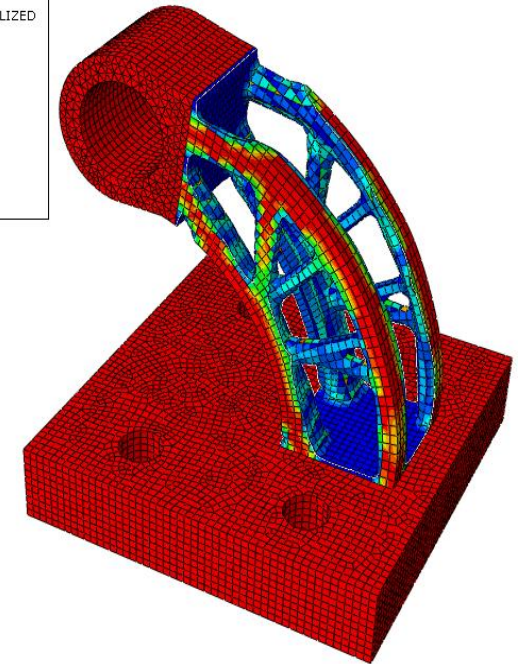
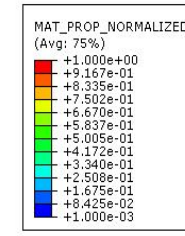
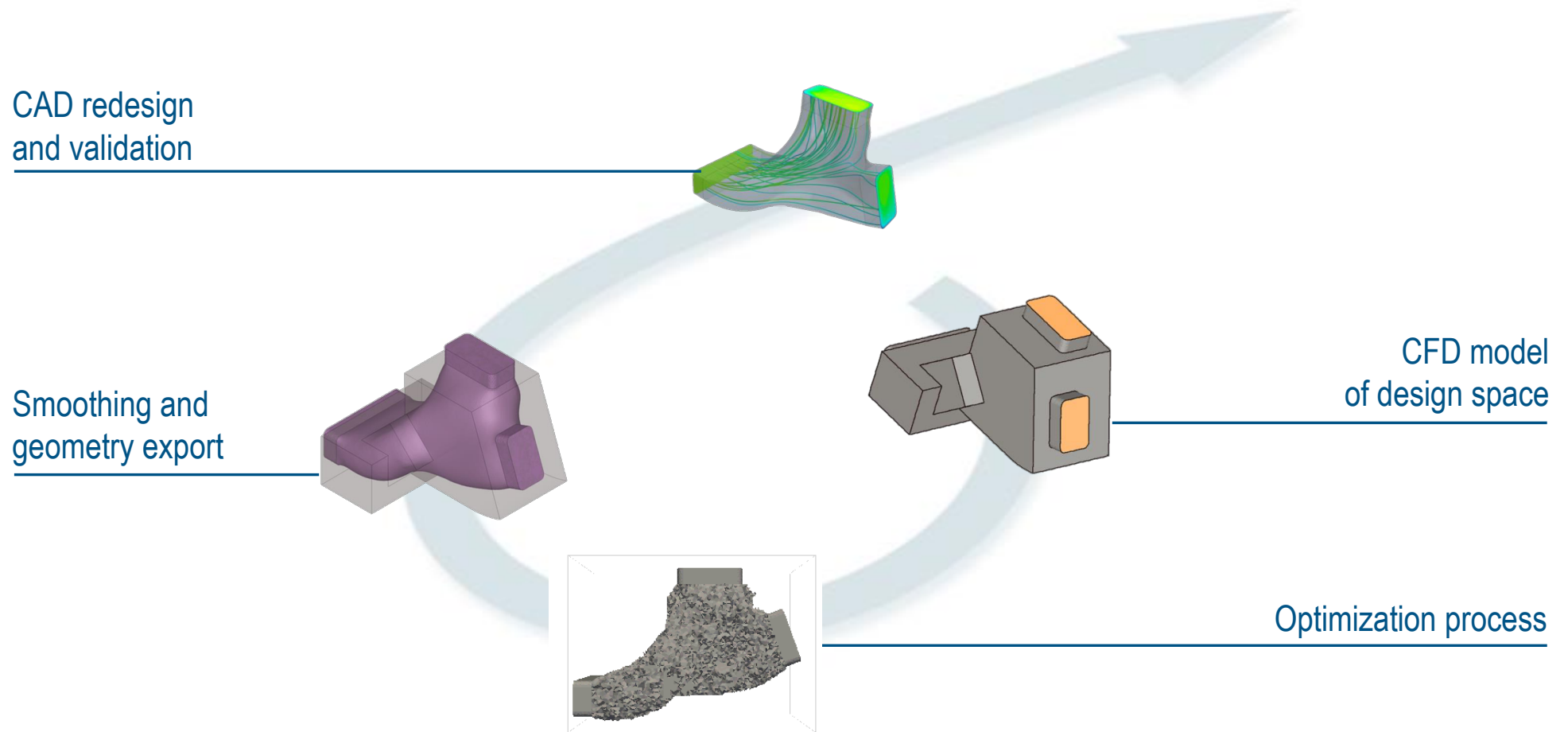


Fig 3 Entire Base Frozen

- We can freeze certain regions to effectively generate structure that can be manufactured effectively.
- The structure in the Figure 3 can be manufactured easily while structure in Fig 2 would have to be manufactured by additive manufacturing.

Fluid Optimization Workflow



谢谢！